

Univariate Calculus & Taylor Series

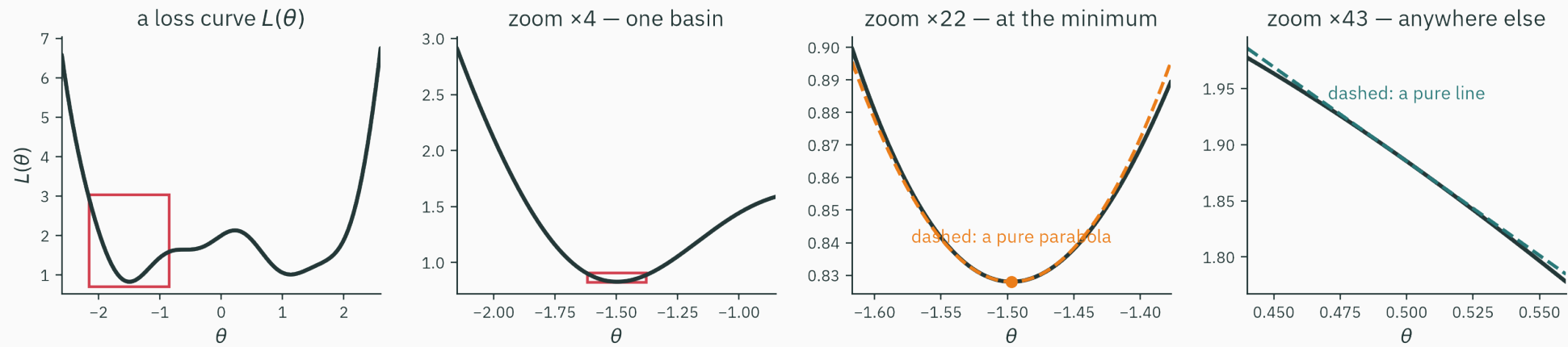
Every smooth function is locally a line

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Mathematical Foundations for AI · IIT Gandhinagar

Local linear and quadratic models of a loss

Zoom in on any loss curve



Wiggly from far away. Zoom at the minimum: a **parabola** appears. Zoom anywhere else: a **line**. Try any smooth loss — same two shapes, every time.

Guess before we continue: why a *line* almost everywhere, but a *parabola* exactly at the bottom? The answer is one formula — you'll derive it today.

Today's one idea

A derivative is the best local linear replacement of a function. Taylor series is the systematic upgrade to polynomial replacements.

Why ML cares: training never sees the whole loss — only a **tiny neighbourhood** of the current weights. Local replacements are the **only** map it has.

Three questions, in order:

1. What is the best local **line**? → the derivative
2. Can we do better than a line? → Taylor's parabola, cubic, ...
3. Where does the local story **break**? → convergence radius

Learning outcomes

By the end of this lecture you will be able to:

1. Read $f'(x_0)$ as a **sensitivity**: Δ_{output} per Δ_{input} — and estimate it from a numeric table.
2. Use the **local line** $f(x_0 + h) \approx f(x_0) + f'(x_0) h$ to predict function values.
3. Apply the **chain rule** as ratio bookkeeping through a composition.
4. Classify flat points with the **second derivative** — curvature and its sign.
5. Build the **Taylor polynomial** of a function and bound its error (★ $\cos x$ at 0).
6. Explain the **convergence radius** — why a Taylor series can fail far from its base point.

A derivative is a sensitivity

Nudge the input, watch the output

Take $f(x) = x^2$ at $x_0 = 3$, so $f(3) = 9$. Nudge x and **measure the response**:

nudge h	$f(3 + h)$	response Δf	ratio $\Delta f/h$
1	16	7	7
0.1	9.61	0.61	6.1
0.01	9.0601	0.0601	6.01
0.001	9.006001	0.006001	6.001

The response shrinks with the nudge — but the **ratio** stabilizes: 7, 6.1, 6.01, 6.001, ...

The ratio is heading somewhere: 6. That destination is the derivative $f'(3)$.

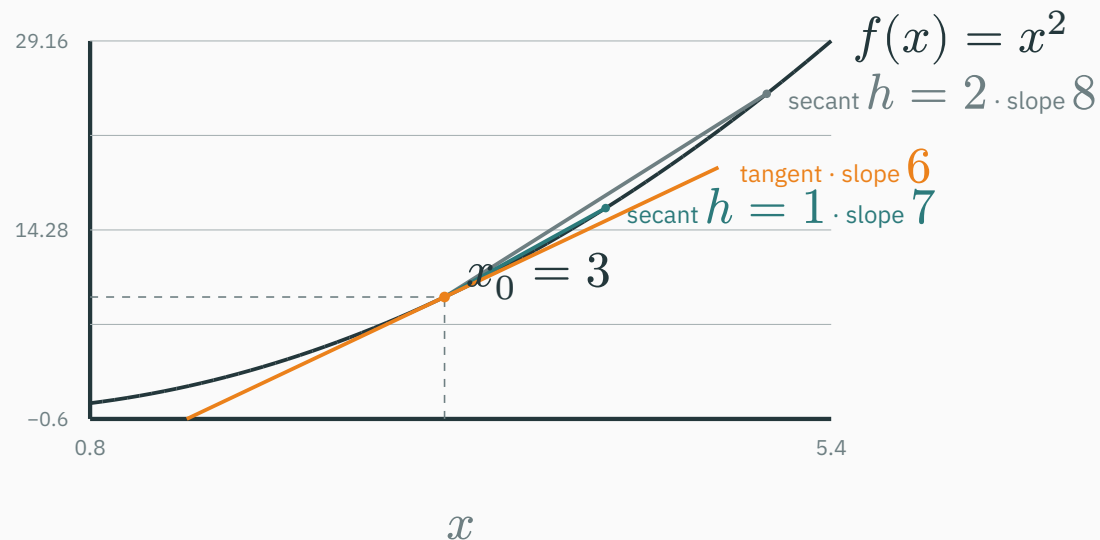
The nudge machine, in code

```
f = lambda x: x**2
for h in [1, 0.1, 0.01, 0.001]:
    print(h, (f(3 + h) - f(3)) / h)
# 1          7.0
# 0.1        6.1000000000000012
# 0.01       6.0099999999999849
# 0.001      6.0009999999999479
```

L2 callback — see the garbage digits (...012, ...849)? That's floating-point rounding leaking into $\Delta f/h$. Shrink h too far and the noise **wins**.

The full war — truncation error vs rounding error in numerical derivatives — is L10's opening act. Today we take the **exact** route instead: limits.

The picture: secants tilting into the tangent



Each row of the table is a **secant line** through $(3, 9)$ and $(3 + h, f(3 + h))$. As $h \rightarrow 0$ the secants tilt into a limiting line: the **tangent**.

Only now, the symbols

The number the ratios approach gets a name:

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

- The fraction is the **table**: response over nudge — an average rate across $[x_0, x_0 + h]$.
- The limit is the **stabilized ratio**: the instantaneous rate *at* x_0 .

Notation zoo, all the same object: $f'(x_0)$, $\frac{df}{dx}(x_0)$, $\dot{x}$ (physics). From L8 on, ∂ joins the zoo.

Read derivatives as sensitivities

$f'(x_0)$ answers one question: **if the input moves a little, how hard does the output move?**

function	f'	meaning
position $\rightarrow$ time	≈ 20 m/s	speed
revenue $\rightarrow$ price	-300 ₹/₹	raise price ₹1, lose ₹300
loss $\rightarrow$ one weight	-0.8	raise w by ε , loss drops 0.8ε

The last row is the **only number training ever consults** about a weight — millions of times per run (L17).

$f'(x_0) \approx \frac{\Delta \text{ output}}{\Delta \text{ input}}$ — a plain ratio of small changes. Everything else in this course is bookkeeping around that ratio.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/derivative-tangent

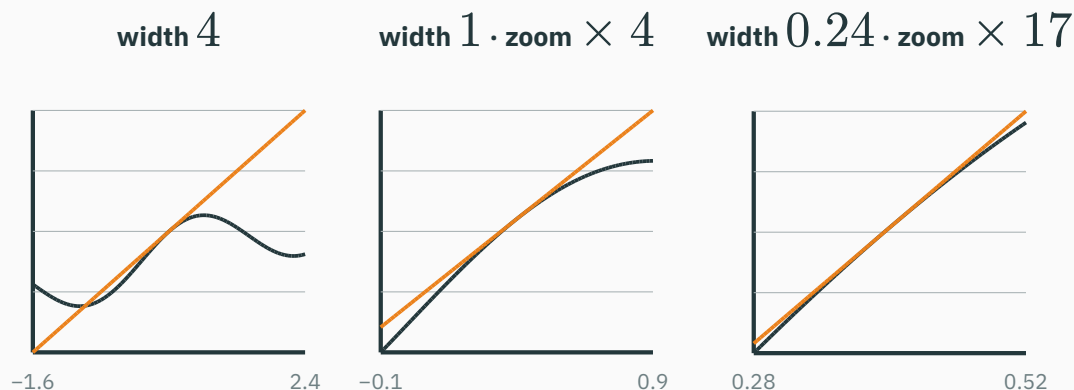
Drag the base point along x^2 , $\sin x$, or $x^3 - x$; shrink the secant step h and watch the average slope converge onto the derivative — then zoom until the curve straightens into its tangent line.

The two sliders are exactly today's two sections: **secant** → **tangent** (the limit) and **zoom** → **straight** (the local line).

A derivative is a local line

Zoom in: every smooth curve straightens

$f(x) = \sin 2x + x/2$ near $x_0 = 0.4$ — same curve, same dashed tangent, three window widths:



At $\text{zoom} \times 17$ the curve and its tangent are **indistinguishable**. “Smooth” *means* this: zoom far enough and there is only one line left.

The local replacement formula

Rearrange the derivative's definition: for small h , the ratio $\approx f'(x_0)$, so

$$f(x_0 + h) \approx \underbrace{f(x_0)}_{\text{where you are}} + \underbrace{f'(x_0)}_{\text{sensitivity}} \cdot h$$

This is a **trade**: swap a function you can't handle for the line you can.

Near x_0 , replace f by the line through $(x_0, f(x_0))$ with slope $f'(x_0)$. That replacement is what “derivative” is *for*.

Test drive: $\sqrt{4.1}$ with no calculator

$f(x) = \sqrt{x}$, base point $x_0 = 4$ (where life is easy): $f(4) = 2$, $f'(x) = 1/(2\sqrt{x})$, so $f'(4) = 0.25$.

$$\sqrt{4.1} \approx 2 + 0.25 \times 0.1 = 2.025$$

Truth: $\sqrt{4.1} = 2.02485\dots$ — the line is off by 0.00015, about **0.008%**, for zero effort.

This is how calculators, game engines, and hand computers before them evaluate functions: keep a few exact anchor points, **replace by lines (or short polynomials) in between.**

What “best line” means

Many lines pass through $(x_0, f(x_0))$. The tangent is the one whose error **dies faster than h** . For $f(x) = x^2$ at 3, the tangent’s error is exactly h^2 :

h	error = $f(3 + h) - (9 + 6h)$	error / h
0.1	0.01	0.1
0.01	0.0001	0.01
0.001	0.000001	0.001

Shrink h by $10 \times \rightarrow$ error shrinks $100 \times$; the **relative** error vanishes.

Pick any other slope $m \neq 6$: the error stays $\approx (6 - m)h$ — proportional to h , never negligible. Only the tangent’s error is superlinear. That is the sense in which it is “best”.

A sensor function satisfies $f(2) = 5$ and $f'(2) = -3$. Best quick estimate of $f(2.1)$?

A. 4.7

B. 5.3

C. 4.97

D. 2.0

Correct: A — $f(2.1) \approx 4.7$

Why: Local line: $f(2 + 0.1) \approx f(2) + f'(2) \times 0.1 = 5 + (-3)(0.1) = 4.7$. B flips the sign of the derivative; C uses $h = 0.01$; D subtracts f' without multiplying by the displacement h .

Rules, and the chain that runs ML

The rules you already know

School calculus, on one slide — this course **refreshes**, it does not reteach:

$(x^n)' = nx^{n-1}$	$(e^x)' = e^x$	$(\ln x)' = 1/x$
$(\sin x)' = \cos x$	$(\cos x)' = -\sin x$	$(af + bg)' = af' + bg'$
$(fg)' = f'g + fg'$	$(f/g)' = \frac{f'g - fg'}{g^2}$	$(f(g(x)))' = f'(g(x))g'(x)$

Only the last cell carries today's weight. Every loss you will ever differentiate is a **composition** — so the chain rule gets its own treatment.

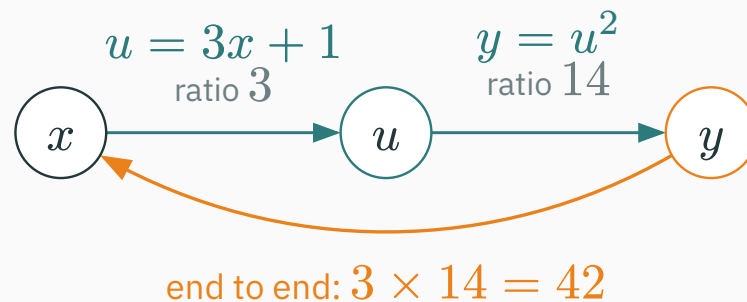
The chain rule is ratio bookkeeping

$y = (3x + 1)^2$ — a two-stage pipeline $x \rightarrow u = 3x + 1 \rightarrow y = u^2$. Nudge $x = 2$ by $\Delta x = 0.01$ and **track the nudge through the pipeline**:

stage	value	nudge arriving	local ratio
x	$2 \rightarrow 2.01$	$\Delta x = 0.01$	—
$u = 3x + 1$	$7 \rightarrow 7.03$	$\Delta u = 0.03$	$\Delta u / \Delta x = 3$
$y = u^2$	$49 \rightarrow 49.4209$	$\Delta y = 0.4209$	$\Delta y / \Delta u = 14.03$

End-to-end sensitivity: $\Delta y / \Delta x = 42.09 \approx 3 \times 14.03$. Each stage **multiplies the nudge by its local ratio** — the bookkeeping is just: multiply the ratios.

The chain rule, in symbols

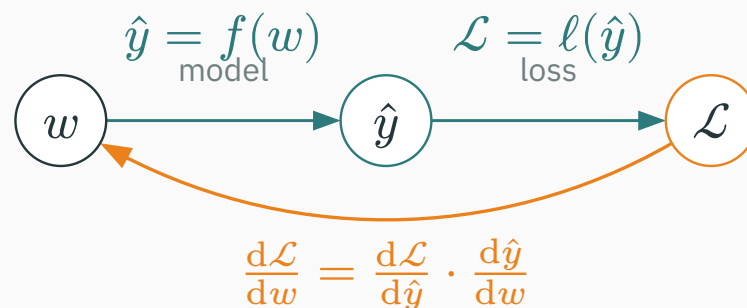


$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad \text{here: } 2u \cdot 3 = 2(7) \cdot 3 = 42$$

In Leibniz notation the du 's **visually cancel** — that's the ratio bookkeeping made typographical.

Evaluate the outer ratio **at the inner value**: $dy/du = 2u$ is read at $u = 7$, not at $x = 2$. Forgetting this is the classic chain-rule bug.

Why ML is one long chain



A real network is this picture **a hundred stages deep**: weights $\rightarrow$ layer $\rightarrow$ layer $\rightarrow$... $\rightarrow$ loss. Training needs $d\mathcal{L}/dw$ for **every** weight.

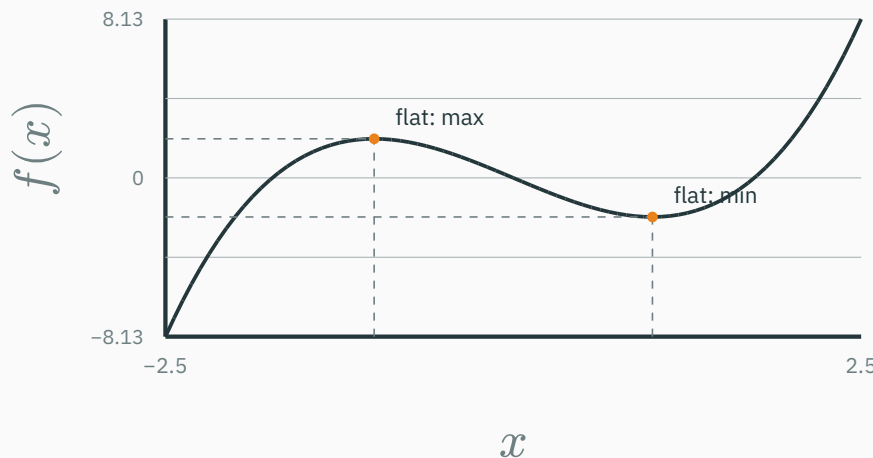
**Same rule, industrial scale: multiply local ratios along the pipeline.
Doing that bookkeeping *efficiently* for millions of weights is
backpropagation — L10–L11.**

Flat points and curvature

Where the local line goes flat

If the best local line has **slope zero**, nudging cannot help you (to first order): x_0 is a **critical point**, candidate max or min. Worked: $f(x) = x^3 - 3x$:

$$f'(x) = 3x^2 - 3 = 0 \Rightarrow x = -1 \text{ or } x = 1$$

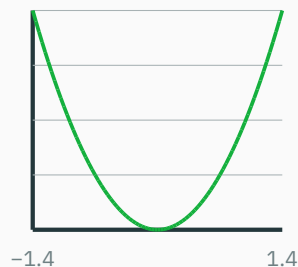


Both flat points have $f' = 0$ — the first derivative **cannot tell them apart**. We need the next order of information.

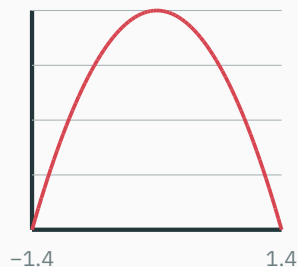
The second derivative is curvature

f'' is the derivative *of the slope*: is the local line **tilting up or down** as you pass through?

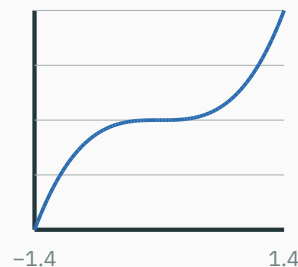
$$f'' > 0 \cdot \text{smile}$$



$$f'' < 0 \cdot \text{frown}$$



$$f'' = 0 \cdot \text{undecided}$$



Sign gallery: **smile** holds water (slope increasing), **frown** spills it (slope decreasing), zero curvature at a point settles nothing by itself.

The second-derivative test

At a critical point ($f'(x_0) = 0$), read the curvature:

curvature	local shape	classification
$f''(x_0) > 0$	smile — curve sits <i>above</i> its flat tangent	local min
$f''(x_0) < 0$	frown — curve sits <i>below</i> its flat tangent	local max
$f''(x_0) = 0$	test is silent	look further

Our worked example: $f''(x) = 6x$, so $f''(-1) = -6 < 0$ (max ✓) and $f''(1) = 6 > 0$ (min ✓) — matching the picture.

Honest footnote: $f(x) = x^4$ has $f'(0) = f''(0) = 0$, yet 0 is a perfectly good minimum. “Silent” means *silent*, not “no extremum”. (L9 upgrades this test to matrices — same fine print.)

Training halts where $\mathcal{L}'(\theta_0) = 0$ and $\mathcal{L}''(\theta_0) = 8$. What does the loss look like locally?

A. a frown — local maximum

B. a smile: $\mathcal{L}(\theta) \approx \mathcal{L}(\theta_0) + 4(\theta - \theta_0)^2$

C. a smile: $\mathcal{L}(\theta) \approx \mathcal{L}(\theta_0) + 8(\theta - \theta_0)^2$

D. flat in every direction — cannot say more

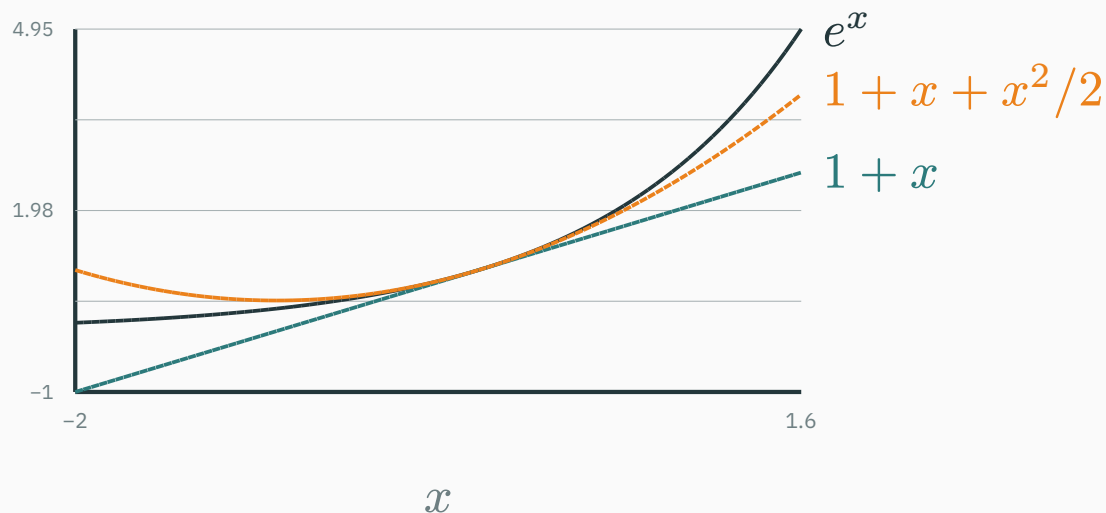
Correct: B — local min, $\mathcal{L}(\theta_0) + 4(\theta - \theta_0)^2$

Why: $\mathcal{L}'' > 0 \rightarrow$ smile $\rightarrow$ local minimum. The parabola's coefficient is $\frac{1}{2}\mathcal{L}''(\theta_0) = 4$, not 8 — that factor $\frac{1}{2}$ is Taylor's, and you are about to see exactly where it comes from.

Taylor: upgrading the line

The line is order one — keep going

The tangent matches f 's **value** and **slope** at x_0 . Next upgrade: also match the **curvature**. $f = e^x$ at 0:



The parabola hugs the curve visibly longer than the line. At $x = 1$: truth 2.718, line 2, parabola 2.5 — each order **halves the miss and more**.

The matching game

Order- n Taylor polynomial: the polynomial that agrees with f at x_0 in value and the first n derivatives — $n + 1$ conditions, $n + 1$ coefficients, one winner.

Why the coefficient of $(x - x_0)^k$ is $f^{(k)}(x_0)/k!$: differentiate $(x - x_0)^k$ exactly k times and the powers rain down:

$$\frac{d^k}{dx^k} (x - x_0)^k = k \cdot (k - 1) \cdot \dots \cdot 1 = k! \quad \Rightarrow \quad \text{coefficient} = f^{(k)} \frac{x_0}{k!}$$

The $k!$ is a **repair factor**: it cancels the rain so the k -th derivatives match exactly.

Your checkpoint's $\frac{1}{2}$ was this factor at $k = 2$: coefficient = $f''(x_0)/2! = f''/2$.

The Taylor formula

$$f(x_0 + h) \approx f(x_0) + f'(x_0) h + \frac{f''(x_0)}{2!} h^2 + \frac{f'''(x_0)}{3!} h^3 + \dots + \frac{f^{(n)}(x_0)}{n!} h^n$$

- Order 0: a constant (stay where you are) · order 1: the local line · order 2: the local parabola...
- Each term is a **correction** the previous order couldn't see.

Near x_0 , Taylor's formula approximates a smooth function by a polynomial whose coefficients are its derivatives at x_0 .

★ Second-order Taylor of $\cos x$, step 1: harvest

D DERIVATION

Everything Taylor needs about $\cos$ at $x_0 = 0$ is three numbers. Harvest them:

$$f(x) = \cos x \quad \rightarrow \quad f(0) = 1$$

$$f'(x) = -\sin x \quad \rightarrow \quad f'(0) = 0$$

$$f''(x) = -\cos x \quad \rightarrow \quad f''(0) = -1$$

Keep harvesting and a pattern locks in: $f'''(0) = \sin 0 = 0$, $f^{(4)}(0) = \cos 0 = 1$, then the cycle $1, 0, -1, 0$ repeats forever. Every *odd* derivative dies at 0 — remember this.

★ Step 2: assemble, then test on numbers

Drop the three harvested numbers into the formula:

$$\cos x \approx 1 + 0 \cdot x + \frac{-1}{2} x^2 = 1 - \frac{x^2}{2}$$

Test at $x = 0.5$, where we can check everything by hand:

$$1 - \frac{0.25}{2} = 0.875 \quad \text{vs} \quad \cos(0.5) = 0.87758... \quad \rightarrow \quad \text{error} = 0.00258$$

Now the suspicious part. The **next** Taylor term would be $x^4/24$, and at $x = 0.5$ that term equals $(0.5)^4/24 = 0.00260$.

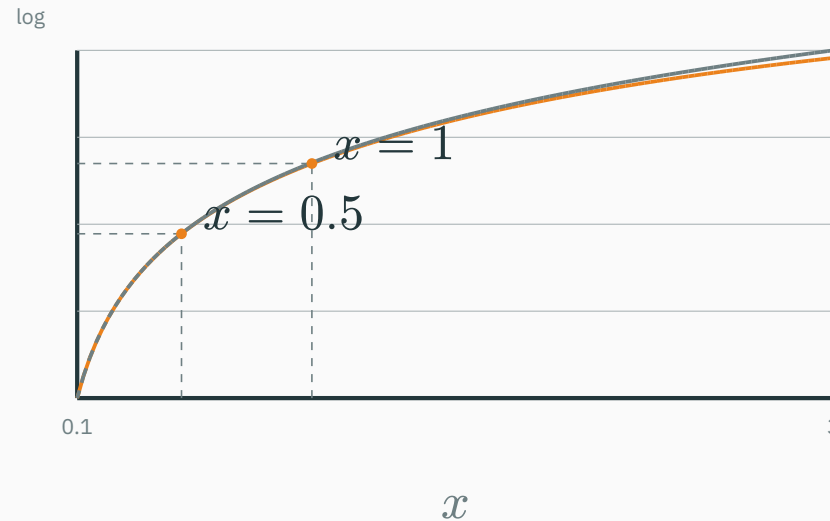
The error **is the first term we dropped, almost exactly. That is Taylor's error story in one number.**

★ Step 3: the error, visualized

D DERIVATION

V VISUAL

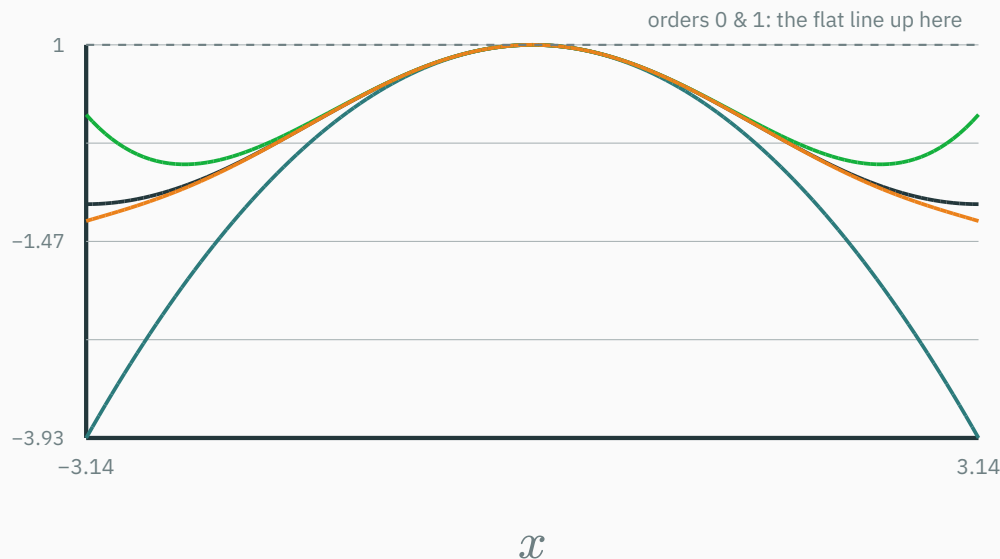
$|\cos x - (1 - x^2/2)|$ against the dropped term $x^4/24$ — log scale, six decades:



error of $1 - x^2/2$ · dashed: $x^4/24$ — glued together until $x \approx 1.5$

Quartic error: halve the distance $x \rightarrow$ the error falls $2^4 = 16 \times$. Second-order replacements are *absurdly* accurate close in.

The full build-up: $\cos x$, order by order



$\cos x$ · orders 0 & 1 · order 2 · order 4 · order 6

Orders 0 and 1 **coincide** — the slope at 0 is zero, so the best line is flat (every odd derivative died in step 1).

Each even order hugs the curve further out: the replacement's **trust region grows** with the order.

The build-up in code

```
from math import cos, factorial
def T(x, n):                                # order-n Taylor of cos at 0
    return sum((-1)**k * x**(2*k) / factorial(2*k)
               for k in range(n // 2 + 1))
for n in [0, 2, 4, 6]:
    print(n, T(1.0, n), abs(cos(1.0) - T(1.0, n)))
# 0  1.0                0.4597
# 2  0.5                0.0403
# 4  0.5416666666666666  0.00136
# 6  0.5402777777777777  2.45e-05
```

At $x = 1$, every two orders buy roughly a factor **30** of accuracy. Six terms of grade-school arithmetic already give $\cos(1)$ to 4 digits — this loop is how libraries compute $\cos$.

The order-2 error of $\cos$ at $x = 0.2$ is about 6.7×10^{-5} . Doubling to $x = 0.4$ makes the error roughly...

A. $2 \times$ bigger

B. $4 \times$ bigger

C. $8 \times$ bigger

D. $16 \times$ bigger

Correct: D — $16 \times$ — **quartic scaling**

Why: Error $\approx x^4/24$, so doubling x multiplies it by $2^4 = 16$. (Check: at 0.4 the true error is $1.06 \times 10^{-3} \approx 15.9 \times$ bigger.) General rule: drop terms above order n , and the error scales like h^{n+1} .

Where the local story breaks

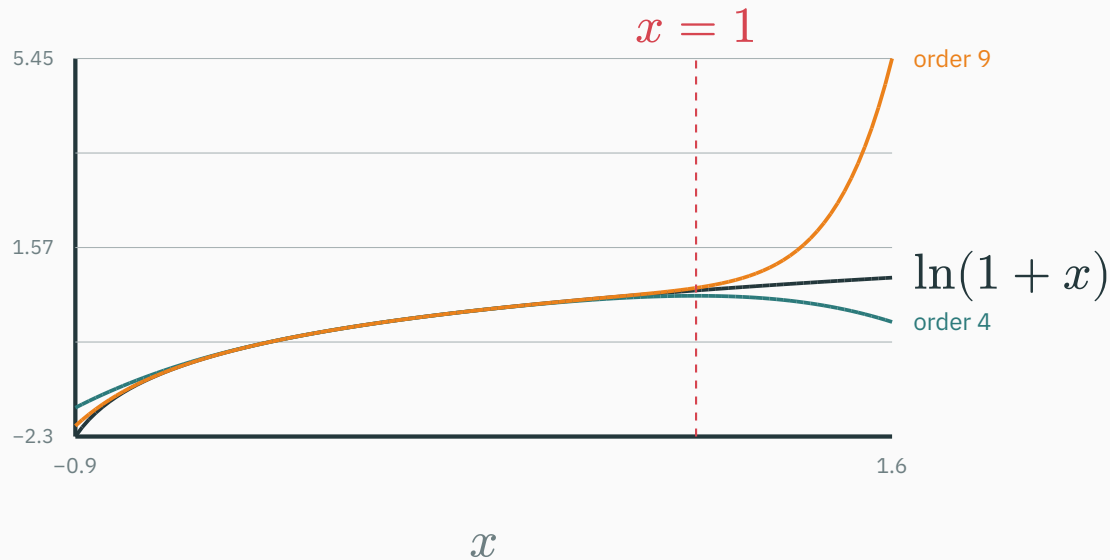
$\log(1 + x)$: a series with a boundary

Same game, new function: $\ln(1 + x)$ at 0 has the series $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$ Watch its partial sums at two inputs:

	S_1	S_2	S_3	S_4	S_5	S_6	truth
$x = 0.5$	0.5	0.375	0.417	0.401	0.407	0.405	$\ln 1.5 = 0.405$ ✓
$x = 2$	2	0	2.67	-1.33	5.07	-5.6	$\ln 3 = 1.099$ never

At $x = 0.5$ the sums settle beautifully. At $x = 2$ they **oscillate harder with every term** — the series diverges, even though $\ln 3$ is a perfectly innocent number.

Watch it break



Inside $|x| < 1$: higher order = better, everywhere. Past $x = 1$ the orders **disagree violently** — order 9 rockets up, order 4 dives — while the truth ambles along.

The radius of convergence

Why 1? The series was built entirely from data **at $x = 0$** — and $\ln(1 + x)$ has a disaster at $x = -1$ (it plunges to $-\infty$).

A Taylor series converges inside a radius equal to the distance from x_0 to the **nearest disaster — and diverges beyond it, in *every* direction.**

The direction doesn't matter! The blow-up lives at -1 , yet the series also fails at $+2$ — a point where the function itself is perfectly finite. The local story simply **cannot see** past its radius.

Intuition to carry: a Taylor series is **local testimony**. Near the base point it is superb evidence; past the radius it is no evidence at all.

And $\cos$ never breaks

$\cos$ has no disaster anywhere — its radius is ∞ . The mechanism: **factorials beat powers**. Terms $x^k/k!$ at the distant point $x = 10$:

k	10	20	30	40
$10^k/k!$	2756	41	0.0038	1.2×10^{-8}

The terms **rise first** — then $k!$ catches up and crushes them. Enough terms deliver $\cos(10)$ to any accuracy you like.

So both endings exist: e^x , $\sin$, $\cos$ converge everywhere; $\ln(1+x)$ and $\frac{1}{1-x}$ have a finite horizon. Always ask *where* a local replacement is licensed — in ML we stay close to x_0 , which is why Taylor serves us so well.

What this buys ML

Apply the quadratic model to the opening loss

Zoom near a minimum θ^* and Taylor-expand the loss there:

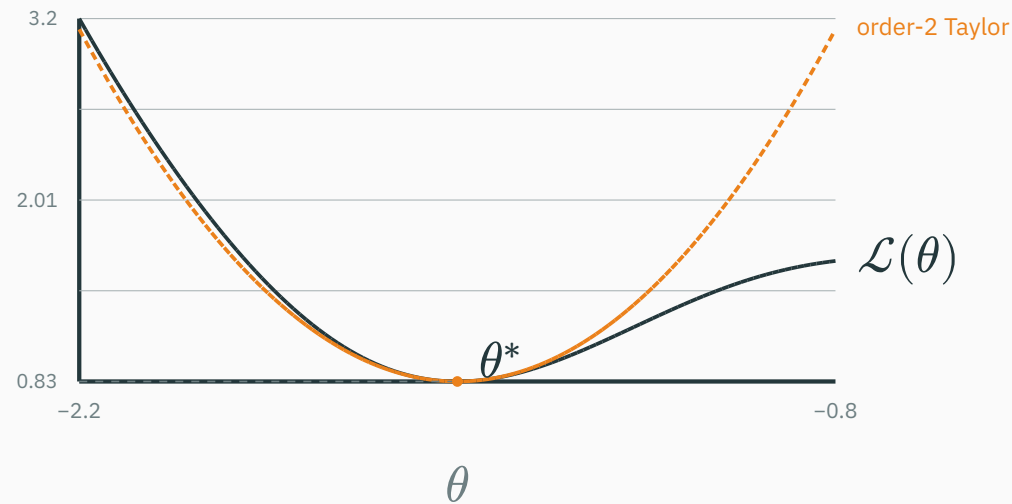
$$\mathcal{L}(\theta) \approx \mathcal{L}(\theta^*) + \underbrace{\mathcal{L}'(\theta^*)}_{=0 \text{ at a minimum}} (\theta - \theta^*) + \frac{1}{2} \mathcal{L}''(\theta^*) (\theta - \theta^*)^2$$

At the bottom, the **line term is dead** — the first surviving correction is the quadratic. That is why every smooth loss bottoms out as a parabola.

Anywhere else, $\mathcal{L}' \neq 0$: the line term dominates all others for small steps — which is why zooming elsewhere gave a **line**.

Wiggly $\rightarrow$ parabola at minima, line elsewhere: not a coincidence — it's Taylor, order by order.

Near its minimum ($\theta^* = -1.497$, $\mathcal{L}^* = 0.828$, $\mathcal{L}''(\theta^*) = 9.39$), compare the opening loss with the quadratic model $\mathcal{L}^* + 4.69 (\theta - \theta^*)^2$:



Near θ^* the parabola is the loss, for every practical purpose. Optimizers exploit exactly this — next slide's promissory note.

A promissory note: where L7 gets cashed

Continuity ledger — **planted today, paid off later:**

lecture	what it does with today's Taylor
L8–L9	slope $\rightarrow$ gradient, curvature $\rightarrow$ Hessian: same formulas, more dimensions
L16	convexity: the smile never flips — one global basin, no bad minima
L17	gradient descent derived : trust the line, step downhill
L18	Newton's method derived : trust the parabola, jump to its bottom

When L17 writes $\theta_{t+1} = \theta_t - \eta \mathcal{L}'(\theta_t)$, it is quoting today's local line; when L18 divides by $\mathcal{L}''$, it is quoting today's parabola.

Summary & what's next

Lecture 7 — summary

- **Derivative = sensitivity**: the ratio $\Delta f/h$ stabilizes — 7, 6.1, 6.01, ... $\rightarrow$ 6.
- **Local line**: $f(x_0 + h) \approx f(x_0) + f'(x_0)h$ — the only error that beats h .
- **Chain rule**: pipelines multiply local ratios ($3 \times 14 = 42$); ML is a deep pipeline.
- **Flat + curvature**: $f' = 0$ finds candidates; the sign of f'' classifies them — usually.
- **Taylor**: match n derivatives, repair by $k!$; ★ $\cos x \approx 1 - x^2/2$, error $\approx x^4/24$.
- **Radius**: local testimony expires at the nearest disaster ($\ln(1+x)$ past $|x| = 1$; $\cos$ never).
- **At a smooth minimum**: the linear term vanishes and the Hessian determines the local quadratic shape used in L16–L18.

Watch: 3b1b *Essence of Calculus* ch. 1–4 + the Taylor video. **Read before L8:** MML §5.1. **T4** (after L8): derivative and chain-rule drills in NumPy.

Practice problems

Try on paper; verify against the T4 notebook.

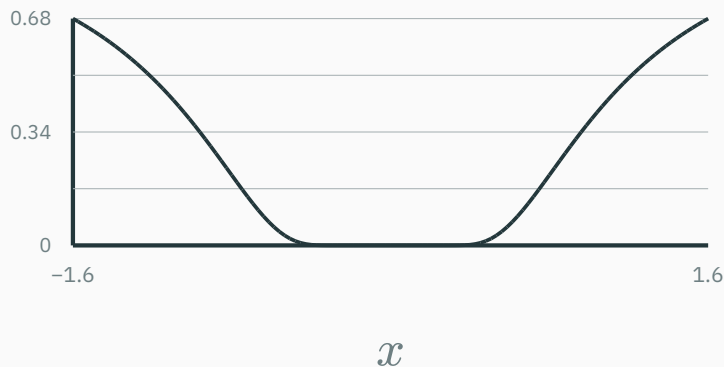
1. Build the nudge table for $f(x) = 1/x$ at $x_0 = 2$ with $h = 0.1, 0.01, 0.001$. What is the ratio heading toward?
2. Estimate $e^{0.1}$ with the local line at 0, then with the order-2 Taylor. Compare both to 1.10517...
3. For $y = (2x - 1)^3$ at $x = 1$: tabulate $\Delta x \rightarrow \Delta u \rightarrow \Delta y$ for $\Delta x = 0.01$ and verify the local ratios multiply to $dy/dx = 6$.
4. Find and classify every critical point of $f(x) = x^3 - 6x^2 + 9x + 1$.
5. Derive the order-3 Taylor of $\sin x$ at 0, evaluate it at $x = 0.3$, and bound the error by the first dropped term. Why do only odd powers appear?
6. The $\ln(1 + x)$ series diverges at $x = 2$ although $\ln 3$ is finite. In one sentence, what sets the radius, and where is the offending point?



The function Taylor cannot see

★ OPTIONAL

$f(x) = e^{-1/x^2}$ (with $f(0) = 0$) is **smooth everywhere** — yet at 0, every derivative equals 0:



Its Taylor series at 0 is $0 + 0x + 0x^2 + \dots \equiv 0$ — it converges beautifully, **to the wrong function**. The curve is flatter at 0 than any polynomial; Taylor is blind to it.

Moral: *smooth* $\neq$ *analytic* (faithfully told by its Taylor series). Our losses never hit the distinction — but honest mathematics keeps the receipt.

Calculus is the art of replacing a function with a line.

Next: **Gradients & the Geometry of Surfaces** — the same replacement idea, in more dimensions: the line becomes a plane, the slope becomes a vector.